

Heart Disease Risk Classifier — End-to-End MLOps Pipeline

Course: AIMLCZG523 — Machine Learning Operations (MLOps), Assignment 01

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Repository: <https://github.com/muthukrishnanram/mlops-assignment1>

Deployed API (local Minikube):

```
minikube start --driver=docker --cpus=4 --memory=8192
docker build -t heart-disease-api:latest .
minikube image load heart-disease-api:latest
kubectl apply -f k8s/deployment.yaml -f k8s/service.yaml
minikube addons enable ingress && kubectl apply -f k8s/ingress.yaml
curl --resolve heart-api.local:80:$(minikube ip) http://heart-api.local/health
```

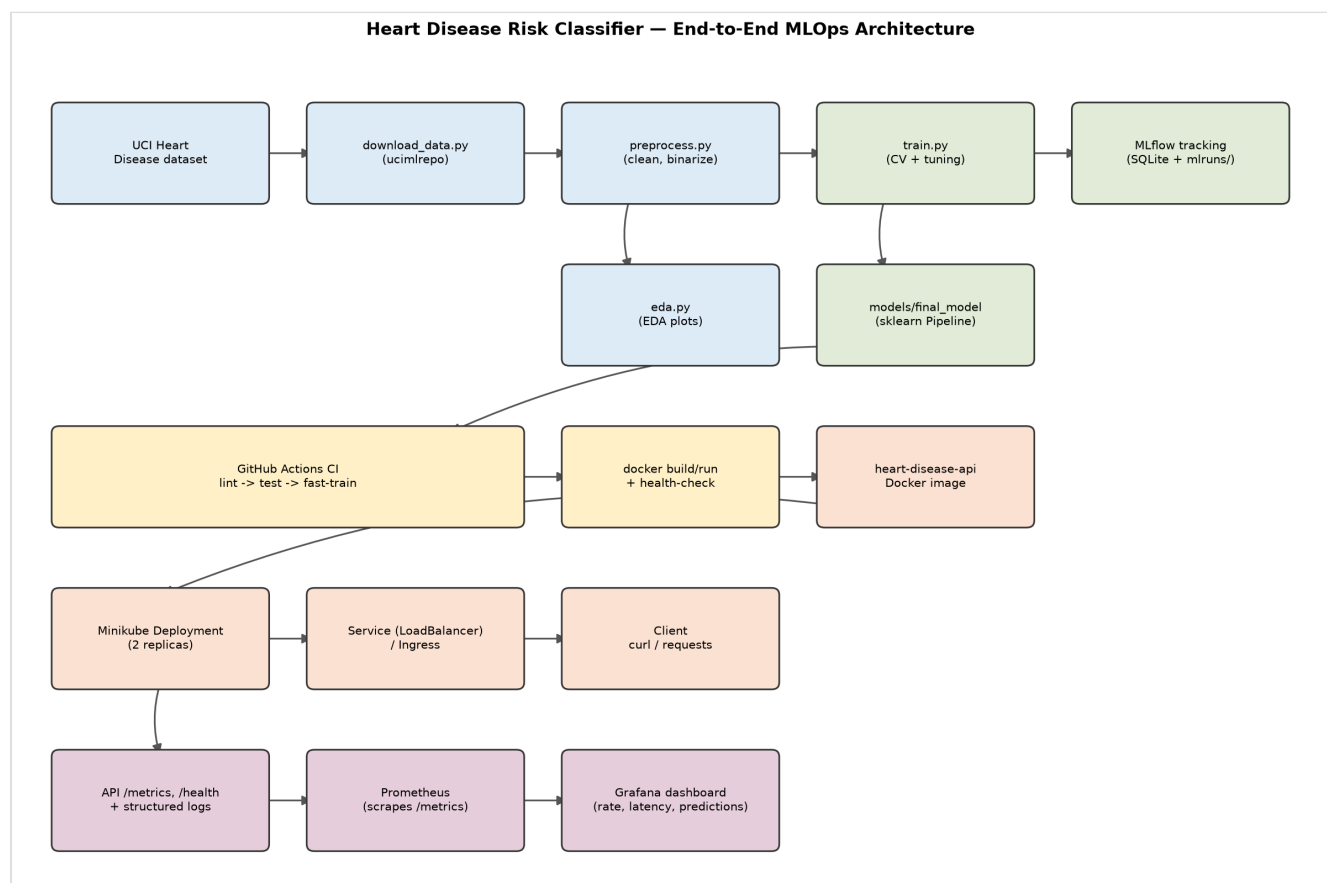
(Full instructions, including the `LoadBalancer` + `minikube tunnel` alternative: `k8s/README.md`.)

1. Executive Summary

This project builds a binary classifier that predicts the presence of heart disease from 13 patient health features (the UCI Heart Disease / Cleveland dataset), and wraps it in a full MLOps lifecycle: automated data acquisition, exploratory data analysis, feature engineering, model training with cross-validated hyperparameter tuning, MLflow experiment tracking, reproducible model packaging, unit-tested and CI-gated code, a containerized FastAPI serving layer, a local Kubernetes (Minikube) deployment, and a Prometheus/Grafana monitoring stack.

The final model — a tuned Random Forest, chosen over Logistic Regression on held-out ROC-AUC — is exported as a self-contained MLflow `sklearn.Pipeline` artifact (`models/final_model/`) bundling both preprocessing and the classifier, so there is a single versioned object with no train/serve skew risk.

2. Architecture



Data flows left to right, top to bottom: the raw dataset is fetched and cleaned, branching into an EDA report and a tuned/tracked training run that exports a packaged model. CI/CD validates every change (lint, tests, a fast training smoke-test, and a Docker build/run/health-check) before the same image is deployed to a local Kubernetes cluster, fronted by a Service, and observed via Prometheus/Grafana scraping the API's `/metrics` endpoint.

3. Setup / Install Instructions

```
git clone <repo-url> && cd mlops-assignment-1
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements-dev.txt      # training/dev/test extras on top of the base deps

python data/download_data.py             # fetch raw UCI data -> data/raw/
python src/data/preprocess.py            # clean -> data/processed/heart_clean.csv
python src/eda.py                        # EDA plots -> report/figures/
python src/models/train.py               # train, tune, track in MLflow, export final model

pytest -q                               # unit tests (18 tests)
uvicorn api.main:app --reload            # local API on http://localhost:8000
```

Or, as a single command matching the assignment's "must execute from clean setup" requirement: `make setup` (see `Makefile`) runs venv creation, dependency install, data download/preprocessing, and training in sequence.

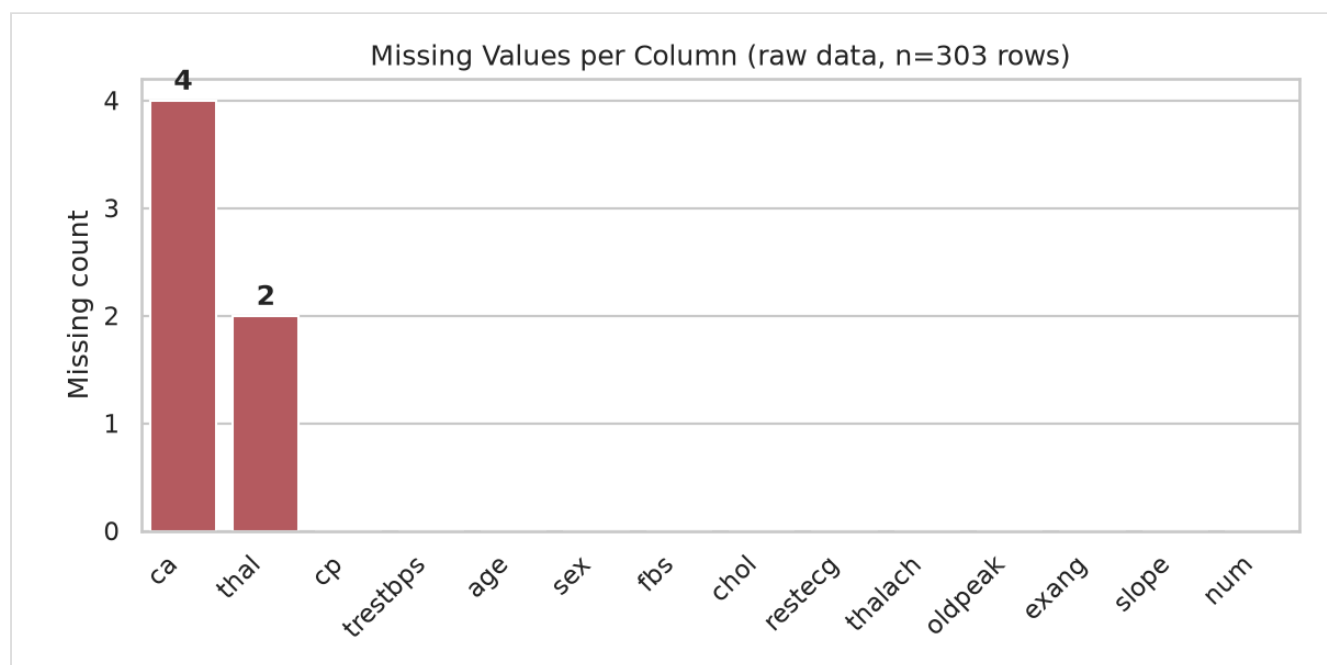
Two requirement files are maintained deliberately separately and compiled together (`scripts/compile_requirements.sh`) so every package they share resolves to an *identical* pinned version in both:

- `requirements.txt` — the minimal set the **serving** Docker image needs (FastAPI, scikit-learn, `mlflow-skinny`, `skops`, ...).
- `requirements-dev.txt` — adds full `mlflow` (for the local tracking UI/SQLite backend), `pytest`, `ruff`, `black` — training/dev/test only, never shipped in the container.

4. Data Acquisition & EDA

Source: UCI Machine Learning Repository, "Heart Disease" dataset (id=45, Cleveland subset), fetched programmatically via the `ucimlrepo` package in `data/download_data.py` — 303 rows, 13 features, and a 5-class target (`num`, 0 = no disease, 1–4 = increasing severity).

Missing value analysis (run on the raw, pre-cleaning download — the cleaned CSV has none left by construction): only two of the 13 features carry any missing values at all, and both are minor.

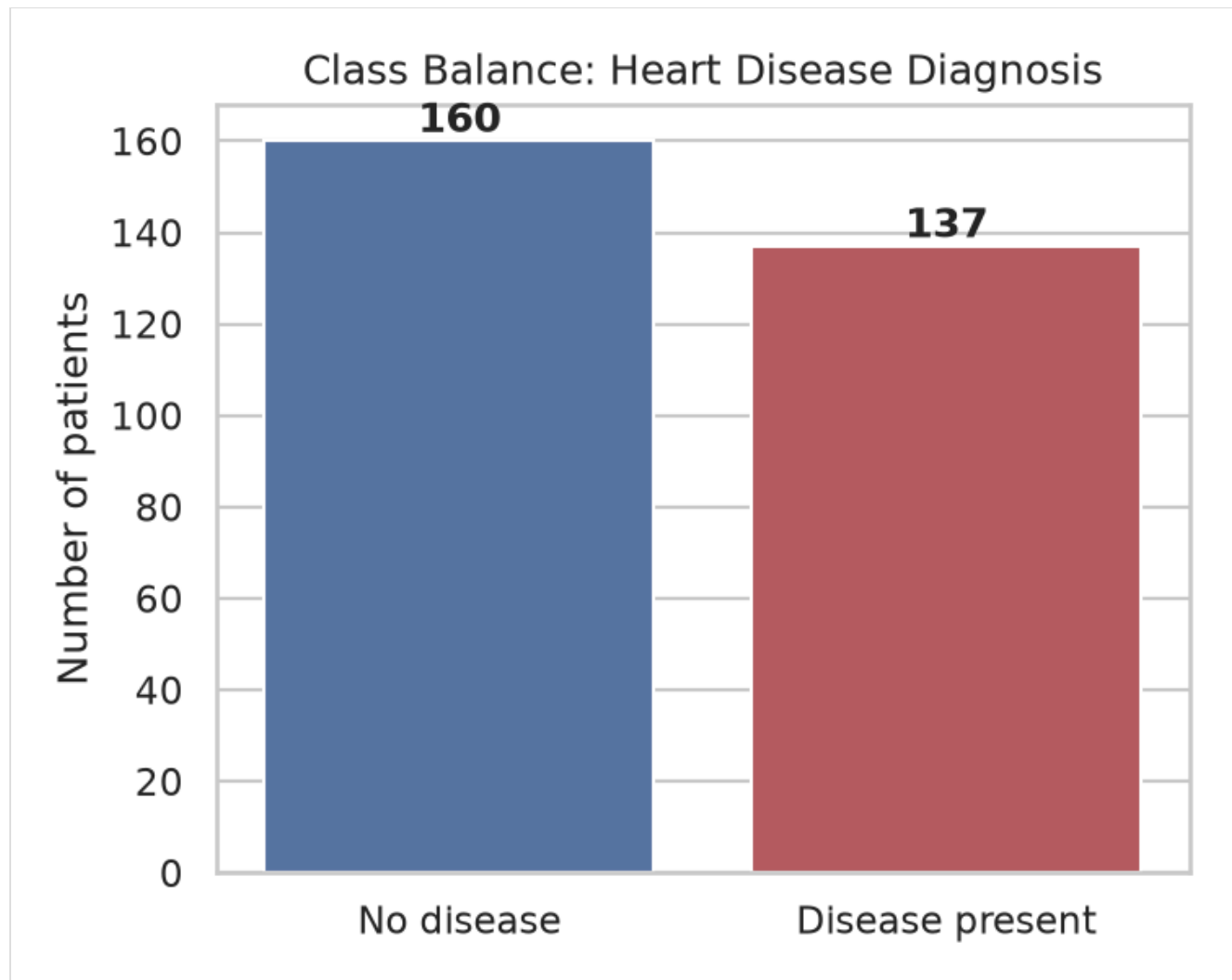


Cleaning (`src/data/preprocess.py`):

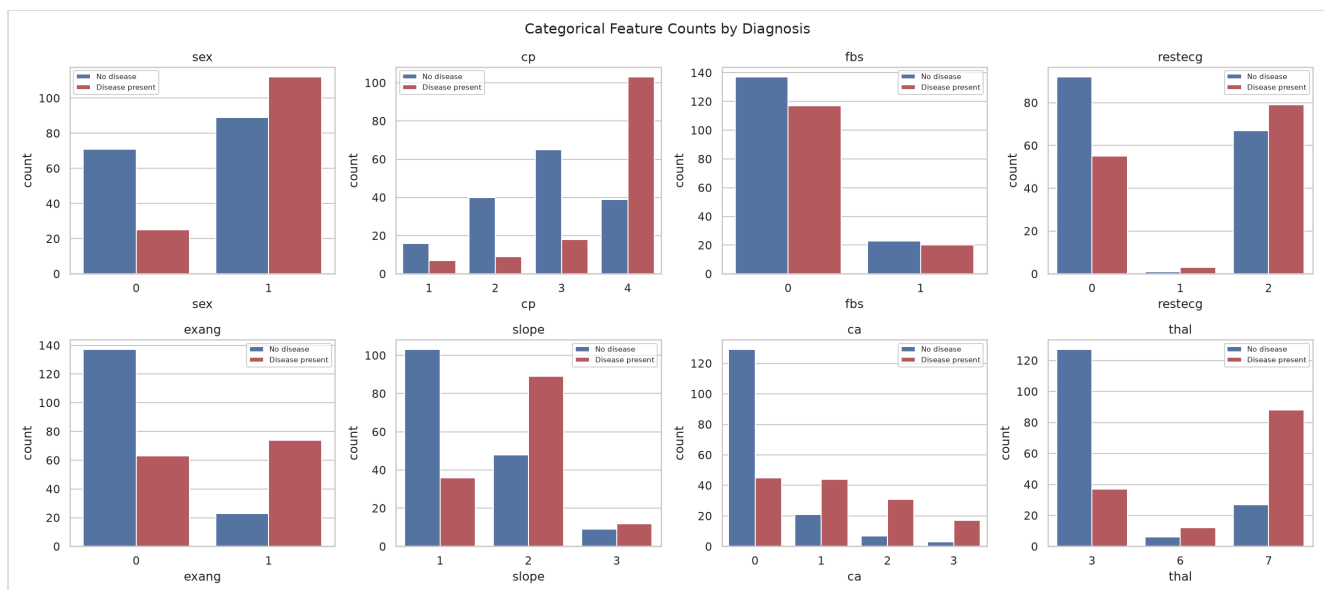
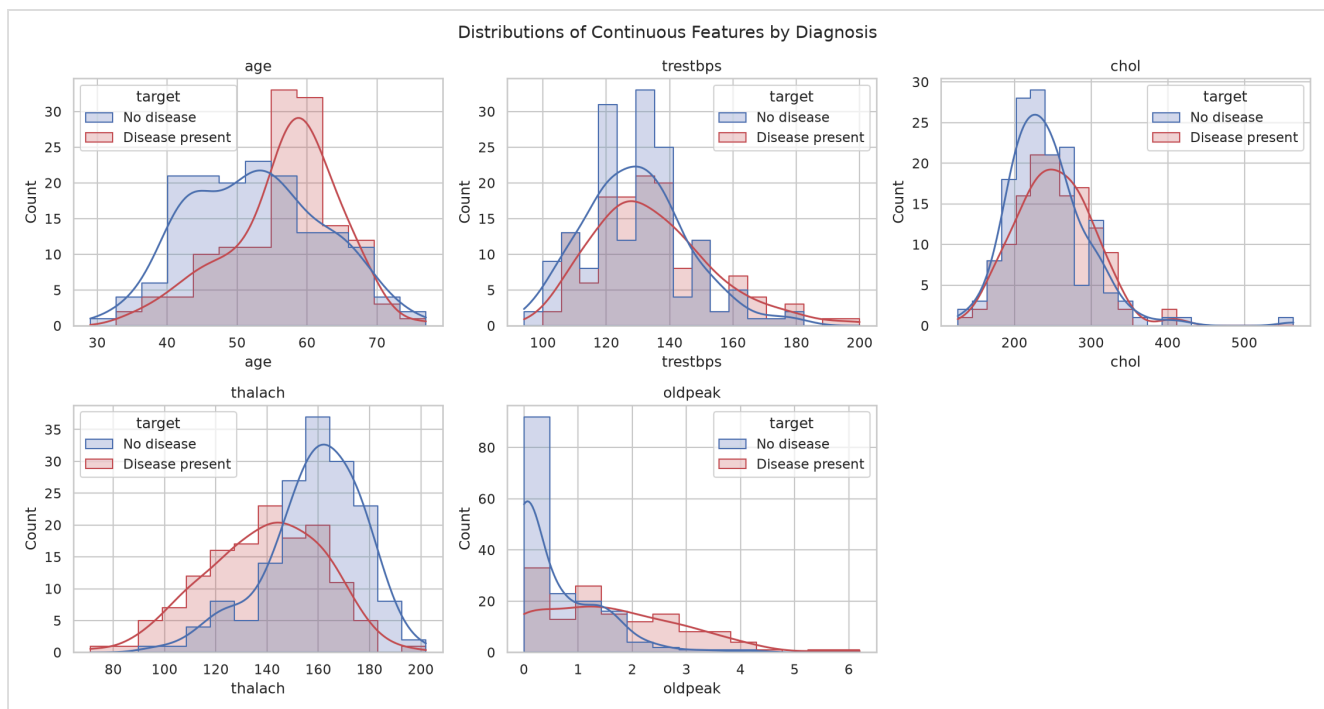
- `ca` and `thal` — two categorical columns — had 4 and 2 missing values respectively (6 of 303 rows total, ~2%). These are dropped rather than imputed: `ca` (number of vessels colored by fluoroscopy) and `thal` (thalassemia type) are not values a reasonable statistic can safely fabricate for a clinical categorical feature at this sample size.

- The 5-class `num` target is binarized to `target` (0 = no disease, 1 = any disease present), matching the assignment's binary classification framing.
- Categorical columns are cast to `int` ; the cleaned set has **297 rows** with **no missing values**, saved to `data/processed/heart_clean.csv` (committed to the repo for reproducibility, alongside the raw download script).

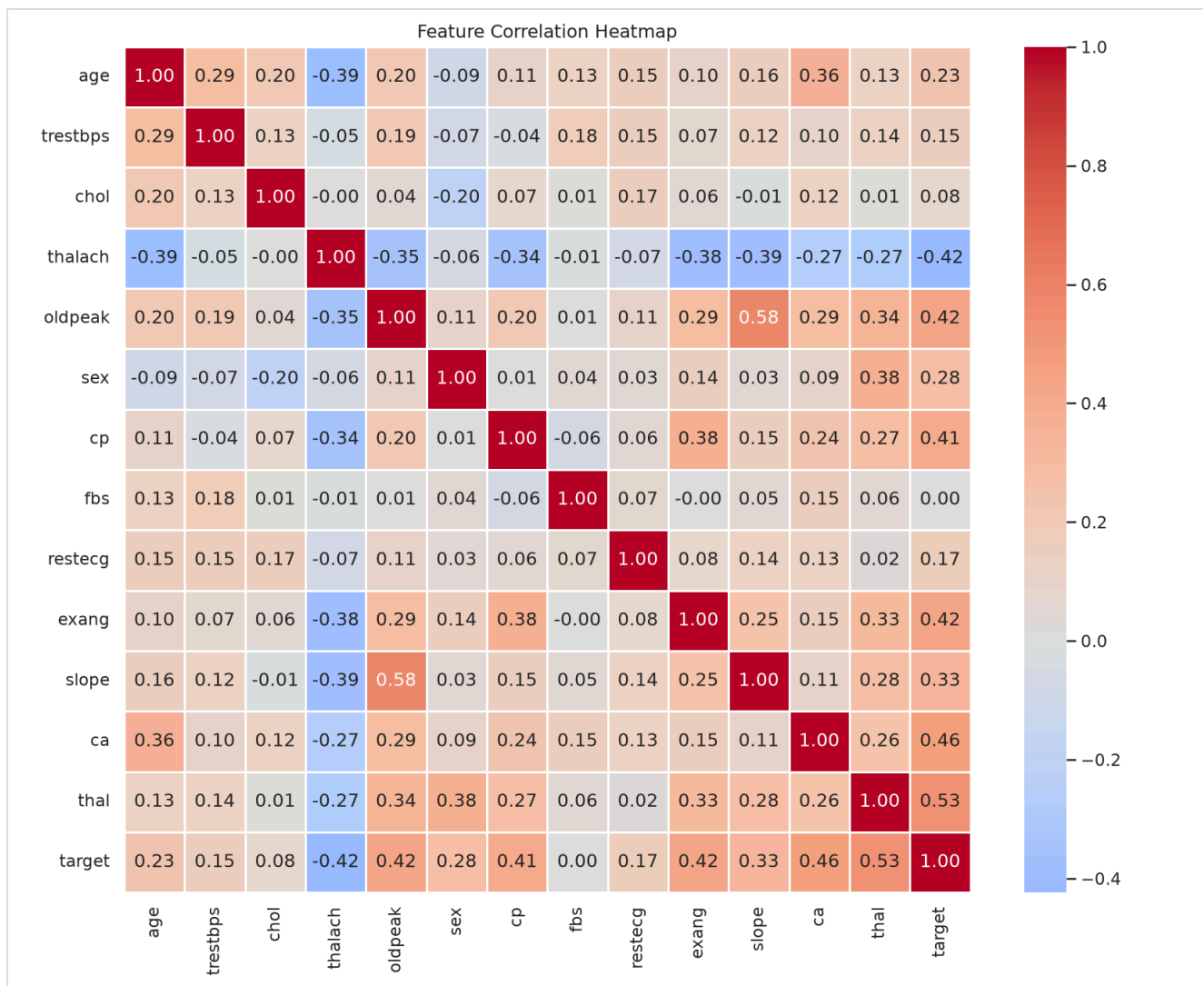
Class balance: 160 "no disease" vs. 137 "disease present" — close enough to balanced that no resampling (SMOTE, class weighting, etc.) was applied.



Feature distributions by diagnosis show several features separating the two classes visibly even before modeling — most notably `thalach` (max heart rate achieved, lower in disease-present patients) and `oldpeak` (ST depression, higher in disease-present patients):



Correlation heatmap (continuous + categorical + target): `thal` (0.53), `ca` (0.46), `oldpeak` (0.42), `exang` (0.42), and `cp` (0.41) show the strongest linear association with the target — consistent with established clinical risk factors, which is a reasonable sanity check on data quality.



Feature relationship analysis — a pairplot of the continuous features (pairwise scatter, diagonal KDE) complements the heatmap's numeric summary with the actual pairwise shapes: `thalach` visibly shifts lower and `oldpeak` higher for disease-present patients across nearly every pairing, and no pair of continuous features shows strong enough collinearity to warrant dropping one (confirming the heatmap's off-diagonal values, all well under 0.6 among the continuous features themselves).



5. Feature Engineering & Model Development

Feature engineering (`src/features/build_features.py`): a single `sklearn.ColumnTransformer` , reused identically by every model:

- **Continuous features** (`age` , `trestbps` , `chol` , `thalach` , `oldpeak`) — `StandardScaler` .
- **Categorical features** (`sex` , `cp` , `fbs` , `restecg` , `exang` , `slope` , `ca` , `thal`) — `OneHotEncoder(drop="if_binary", handle_unknown="ignore")` . Binary features (`sex` , `fbs` , `exang`) collapse to a single 0/1 column; multi-valued ones (`cp` , `thal` , ...) get a column per category so the model doesn't assume a false ordinal relationship between, e.g., chest-pain types. `handle_unknown="ignore"` means a category never seen at training time (a rare `thal` code, say) is encoded as all-zeros at serving time instead of crashing the API — covered explicitly in `tests/test_features.py` .

Models (`src/models/train.py`): three classifiers — two more than the assignment's minimum, and XGBoost is explicitly named as an example choice in the FAQ — each wrapped in one `Pipeline([("preprocessor", ...), ("classifier", ...)])` so preprocessing and model are always versioned and served together:

- **Logistic Regression** — `C` swept over `[0.01, 0.1, 1.0, 10.0]`.
- **Random Forest** — `n_estimators` over `[100, 200, 300]`, `max_depth` over `[None, 5, 10]`, `min_samples_leaf` over `[1, 2, 4]` (27 combinations).
- **XGBoost** — `n_estimators` over `[100, 200, 300]`, `max_depth` over `[3, 5, 7]`, `learning_rate` over `[0.01, 0.1, 0.2]` (27 combinations, for parity with Random Forest's grid size).

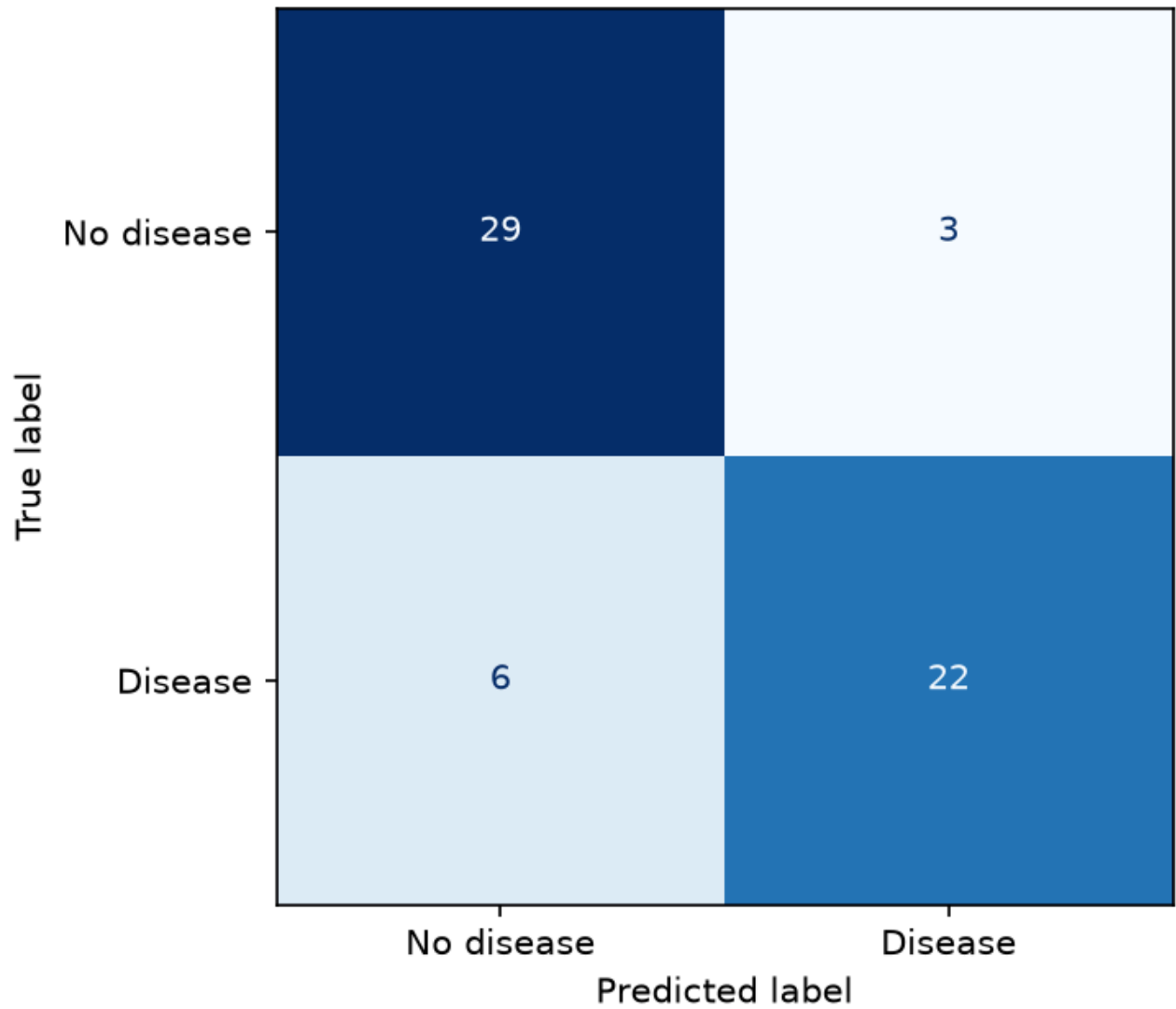
Tuning & evaluation: `GridSearchCV` over `StratifiedKfold(n_splits=5, shuffle=True, random_state=42)`, scoring accuracy/precision/recall/F1/ ROC-AUC simultaneously and refitting on ROC-AUC (the most informative single metric for a moderately-imbalanced binary clinical classifier — it's threshold-independent, unlike accuracy). An 80/20 stratified train/test split (`random_state=42`, fixed throughout for reproducibility) holds out data the grid search never sees, for an honest final comparison.

Results:

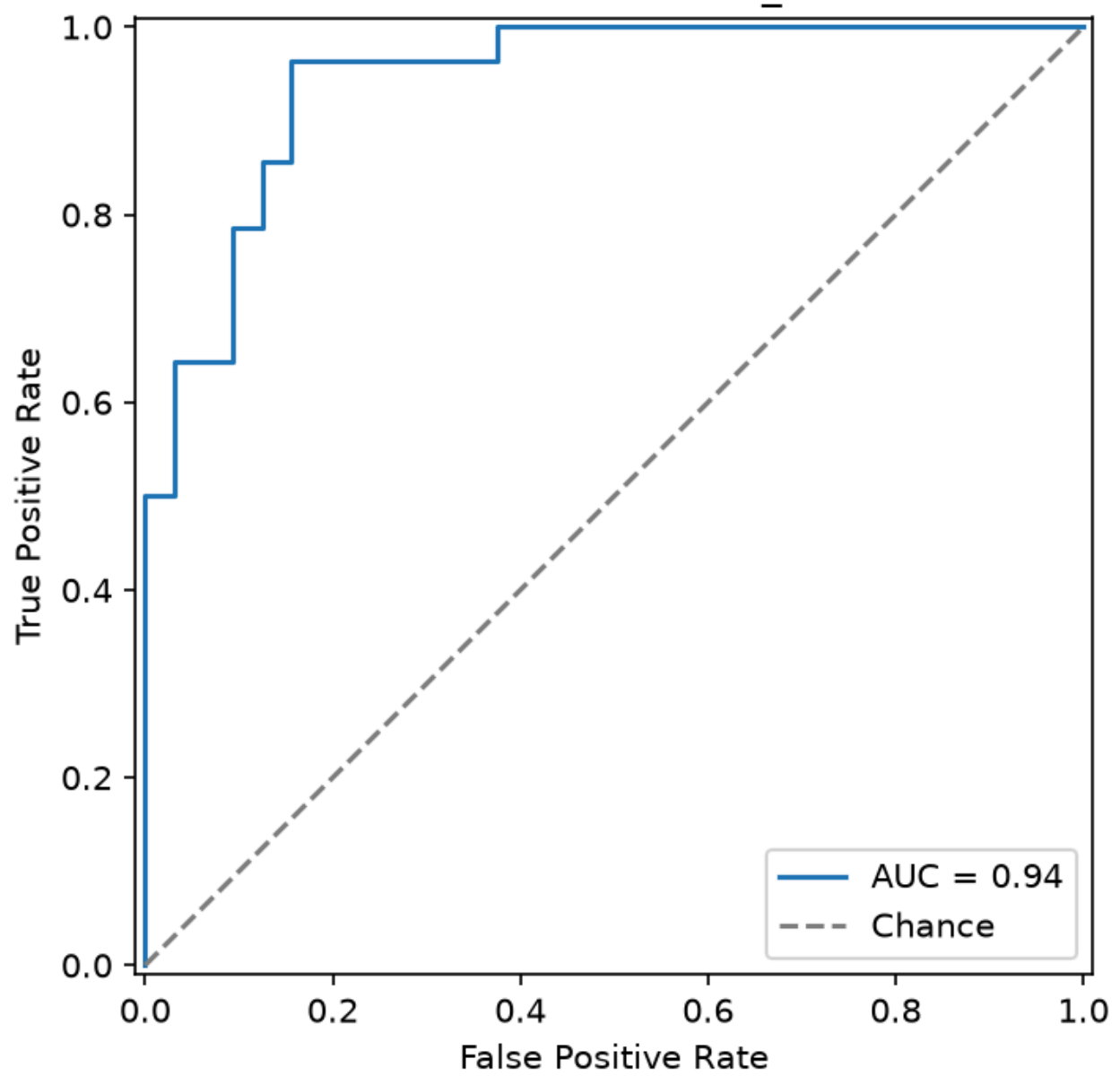
Model	Best params	CV ROC-AUC (mean)	Test accuracy	Test precision	Test recall	Test F1	Test ROC-AUC
Logistic Regression	<code>C=1.0</code>	0.902	0.817	0.870	0.714	0.784	0.938
Random Forest	<code>n_estimators=300,</code> <code>max_depth=None,</code> <code>min_samples_leaf=4</code>	0.906	0.850	0.880	0.786	0.830	0.943
XGBoost	<code>n_estimators=300,</code> <code>max_depth=3,</code> <code>learning_rate=0.01</code>	0.887	0.817	0.870	0.714	0.784	0.921

Random Forest wins narrowly on every held-out metric and is exported as the final model. XGBoost comes in last of the three, not first — on a dataset this small (237 training rows), boosting's extra capacity is a liability more than an asset: the grid search's own winning hyperparameters (`learning_rate=0.01`, the smallest offered) show it pulling *toward* the conservative end of its own search space to avoid overfitting, and it still trails a plain bagged Random Forest and even Logistic Regression on ranking quality (ROC-AUC). This is a legitimate, reportable result, not a tuning failure — it's a reasonable demonstration that "more sophisticated model" doesn't automatically mean "better," especially at this sample size. The gap between Random Forest and Logistic Regression is small enough that Logistic Regression remains a reasonable, more-interpretable fallback — noted here rather than discarded, since interpretability matters for a clinical-adjacent use case.

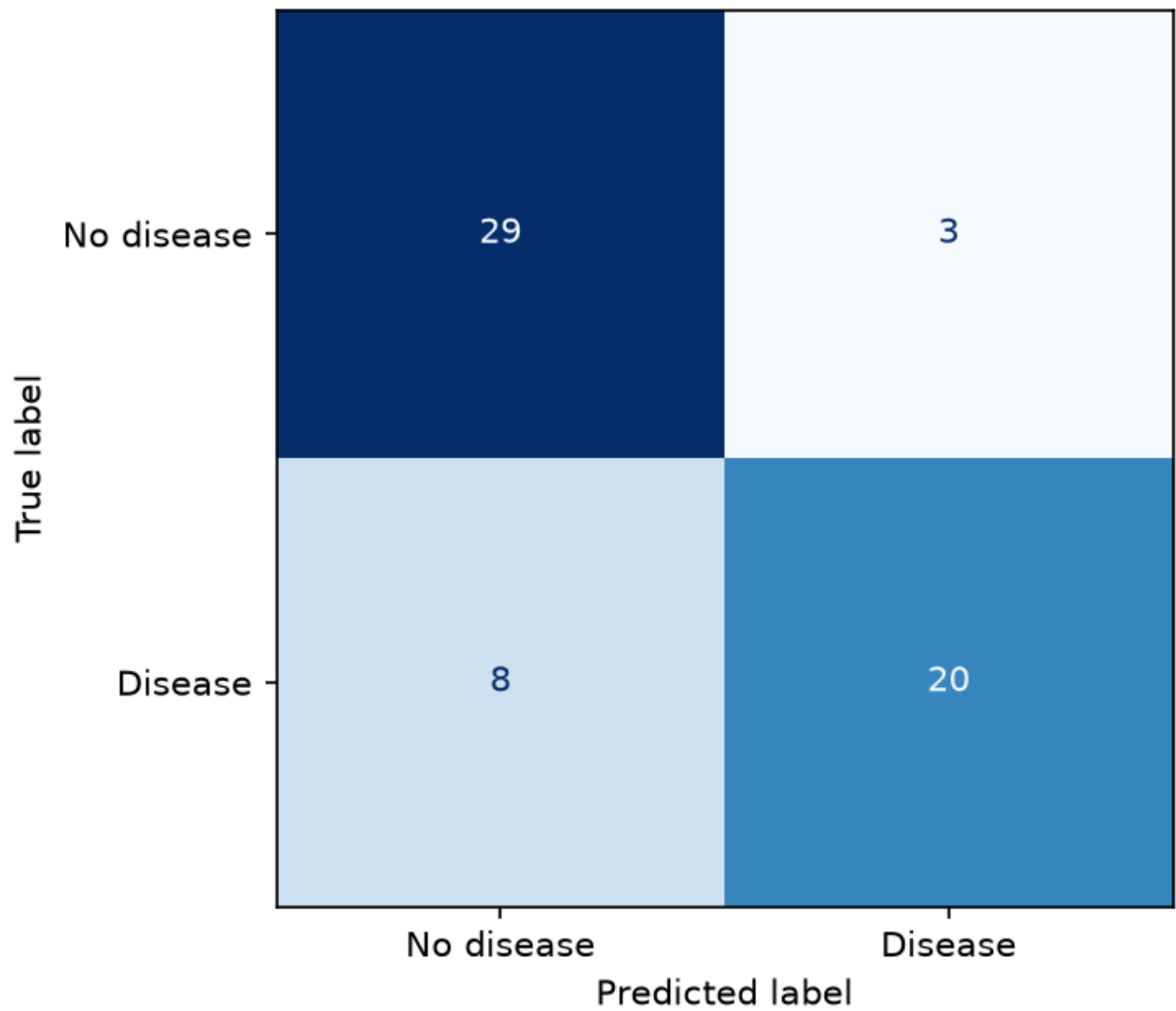
Confusion Matrix — random_forest

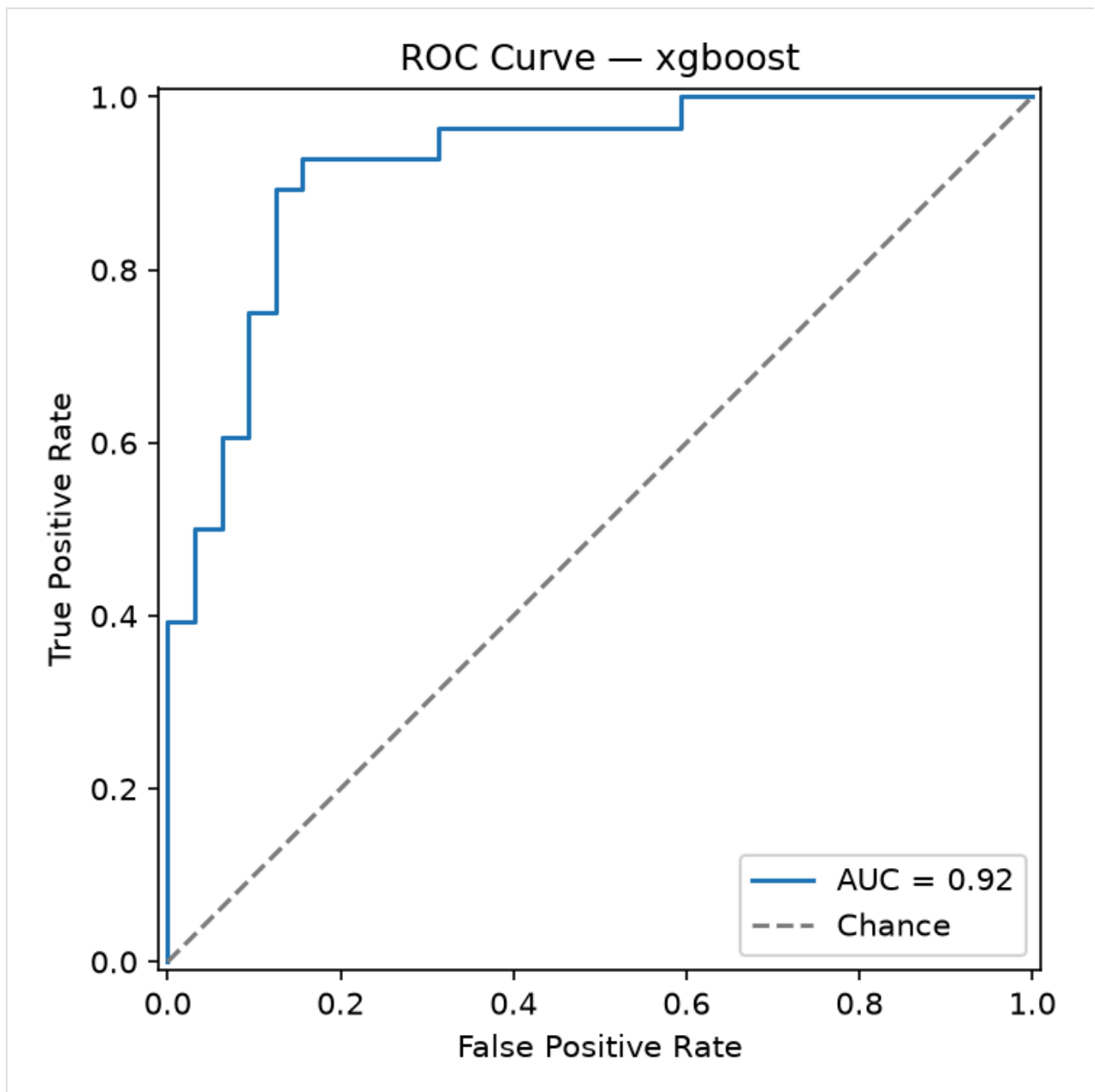


ROC Curve — random_forest



Confusion Matrix — xgboost





`train.py --fast` runs the same pipeline with a 1-combination grid and 2-fold CV (~35s total for all three models) — used only by CI as a smoke test that the training code path still runs end-to-end; the tuned numbers above come from the full run (`python src/models/train.py`, no flags).

6. Experiment Tracking (MLflow)

Every `GridSearchCV` run is logged as one MLflow run under the `heart-disease-classification` experiment: best hyperparameters, mean/std of every CV metric, held-out test metrics, the confusion matrix and ROC curve plots as artifacts, and the fitted pipeline itself (`mlflow.sklearn.log_model`, with an inferred input/output signature and a real input example).

Tracking backend is local SQLite (`mlflow.db`) rather than the plain filesystem store — MLflow 3.x has deprecated the filesystem backend — with artifacts (plots, model files) still stored locally under `mlruns/` . Inspect with:

```
mlflow ui --backend-store-uri sqlite:///mlflow.db
```

The screenshot displays the MLflow web interface. On the left is a sidebar with navigation options: Overview, Training runs (selected), Observability, Traces, Sessions, Evaluation, Judges, Review, Datasets, Evaluation runs, Prompts & versions, Playground, Prompts, Agent versions, AI Gateway, Assistant (Beta), Docs, and Settings. The main panel shows the 'heart-disease-classification' experiment with a 'Training runs' tab. A search bar contains the query 'metrics.rmse < 1 and params.model = "tree"'. Below the search bar are filters for 'All time', 'State: Active', 'Datasets', and 'Sort: Created'. A table lists three training runs:

☐	Run Name	Created	Dataset	Duration	Source	Models
☐	xgboost	6 minutes ago	-	9.2s	train.py	model
☐	random_forest	6 minutes ago	-	11.0s	train.py	model
☐	logistic_regression	6 minutes ago	-	10.3s	train.py	model

At the bottom of the table, it says '3 matching runs'.

All three runs are visible under the `heart-disease-classification` experiment with their source (`train.py`), duration, and linked model artifact. Opening a run shows all 15 logged metrics (5 CV mean/std pairs + 5 held-out test metrics) alongside its parameters and run metadata:

heart-disease-classification > Runs > **random_forest**

Overview Model metrics System metrics Traces Artifacts

Description [No description](#)

Metrics (15)

Metric	Value	Models
cv_accuracy_mean	0.8226950354609928	model
cv_accuracy_std	0.06269742722725112	model
cv_precision_mean	0.8481848739495799	model
cv_precision_std	0.09859382209614219	model
cv_recall_mean	0.7692640692640692	model
cv_recall_std	0.1083341442077329	model
cv_f1_mean	0.79826358039124	model
cv_f1_std	0.06876984139052067	model
cv_roc_auc_mean	0.9056170496170497	model
cv_roc_auc_std	0.05822221408413785	model
test_accuracy	0.85	model
test_precision	0.88	model
test_recall	0.7857142857142857	model
test_f1	0.8301886792452831	model

Parameters (5)

Parameter	Value
model_type	random_forest

About this run

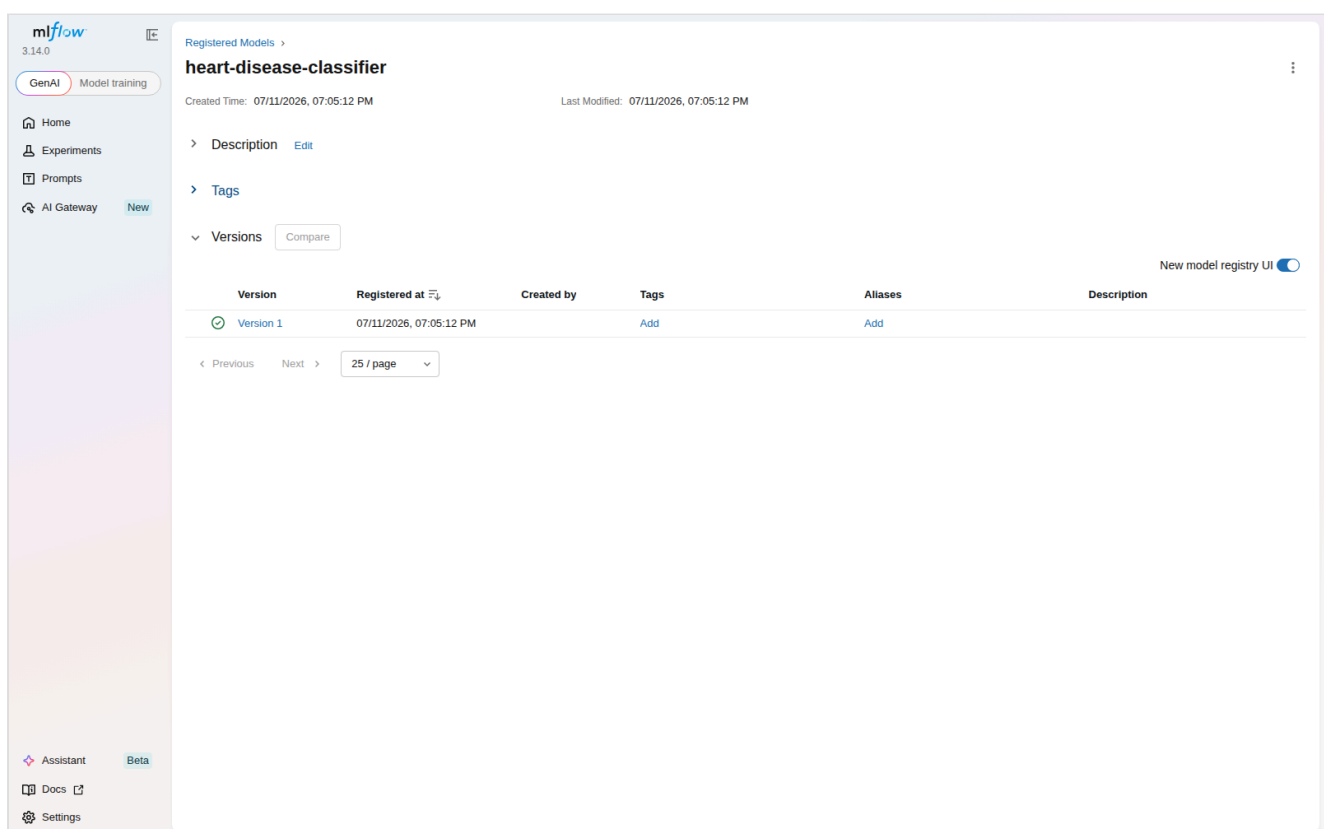
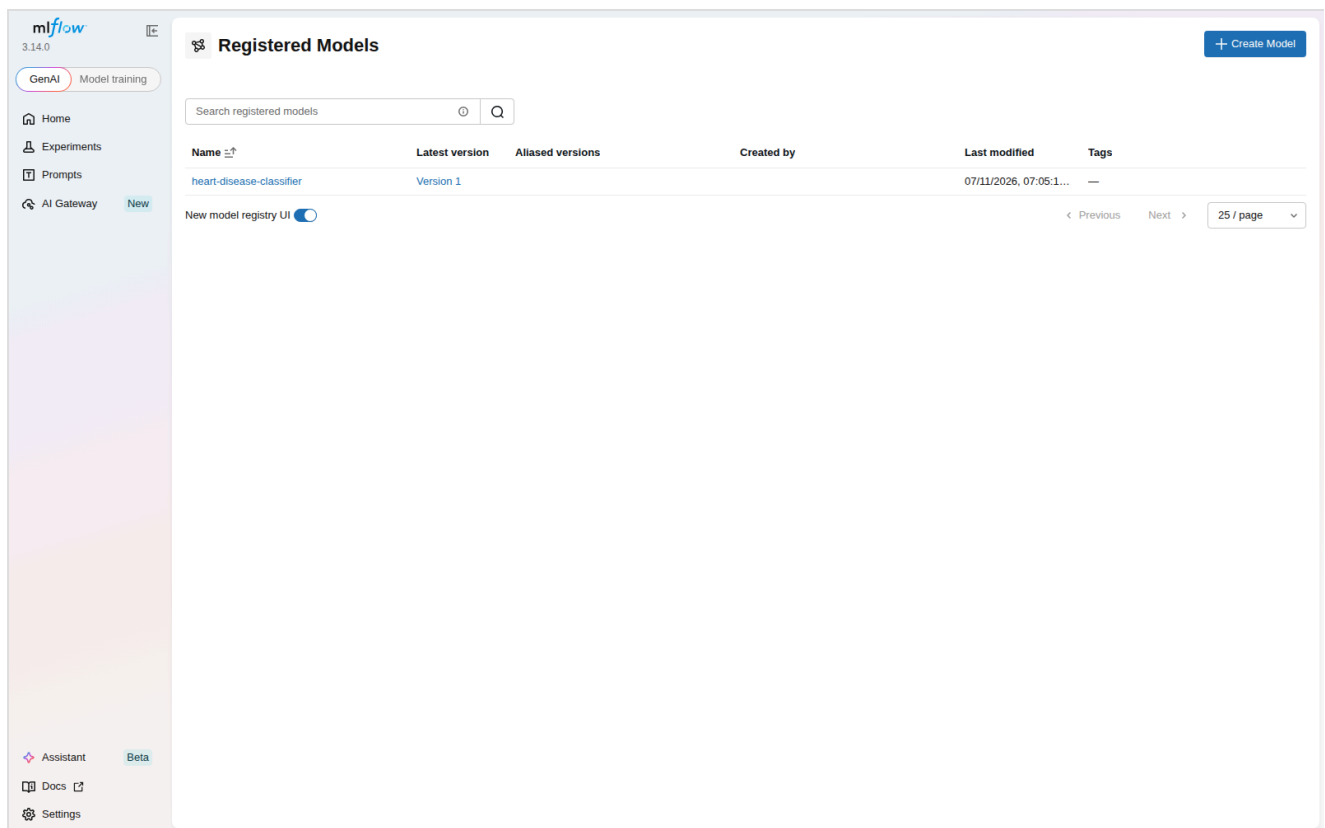
Created at 07/11/2026, 06:25:43 PM
 Created by [krishna](#)
 Experiment ID [1](#)
 Status Finished
 Run ID [764f4c46deb141a4ad8dbde6dad91b31](#)
 Duration 11.0s
 Source [train.py](#) [main](#) [79bd72e](#)
 Registered prompts —

Datasets
None

Tags
[Add tags](#)

Registered models
None

Model Registry: beyond the per-run model artifact above, the winning run's model is also registered as a named, versioned entry — `mlflow.register_model(model_uri=f"runs://{best_run_id}/model", name="heart-disease-classifier")` — creating `heart-disease-classifier` version 1. Every future retrain that produces a new winner adds another version under the same name, giving a versioned history independent of any single run, which is the actual MLflow feature "model versions" refers to (as distinct from just logging a model per run, which was already happening).



`report/metrics_summary.json` also captures a machine-readable summary of all three runs (params, CV mean ROC-AUC, full test metrics) as a durable, diff-able record independent of the `mlruns/` directory (which is gitignored — it's bulky, pickled-per-run, and not meant to be shipped).

7. Model Packaging & Reproducibility

The winning pipeline is exported via `mlflow.sklearn.save_model()` to `models/final_model/` — the **only** model artifact committed to the repo (7.5 MB). It contains:

- `model.skops` — the serialized pipeline, in [skops](#) format rather than raw `pickle`. `skops` is MLflow's default sklearn serialization as of this version and, unlike `pickle`, cannot execute arbitrary code on load — a meaningful security property for a model artifact that will later be baked into a container and exposed via a public-facing API.
- `MLmodel`, `conda.yaml`, `python_env.yaml`, `requirements.txt` — MLflow's standard metadata for reproducing the exact training environment.
- `input_example.json` / `serving_input_example.json` — a real sample row, useful for smoke-testing the loaded model.

Reproducibility was verified directly, not assumed: a throwaway venv was built from *only* `requirements.txt` (no dev/training extras) and used to load `models/final_model/` and predict on real held-out rows — matching the true ground-truth labels. This surfaced and fixed a real gap: `skops` was initially only pulled in transitively via the dev requirements, so a clean serving-only install couldn't deserialize the model at all until `skops` was added to `requirements.in` directly (see `src/models/predict.py` and the commit history for the fix). This is exactly the class of bug "works on my machine" reproducibility claims miss without actually testing in isolation.

Adding XGBoost surfaced a second, related issue: `skops` maintains an allowlist of types it trusts by default, and `xgboost.sklearn.XGBClassifier` / `xgboost.core.Booster` aren't on it — `mlflow.sklearn.save_model()` raised `UntrustedTypesFoundException` the first time an XGBoost pipeline was saved. This is `skops` working as intended (refusing to silently trust arbitrary types), not a bug to route around quietly. Since these are our own freshly-trained model objects, not a third-party file of unknown provenance, explicitly passing `skops_trusted_types=["xgboost.core.Booster", "xgboost.sklearn.XGBClassifier"]` to `save_model()` / `log_model()` is the correct fix (`src/models/train.py`) — verified with a full save → load → predict round trip on a real XGBoost pipeline before trusting it in the actual training run.

8. CI/CD Pipeline & Automated Testing

Unit tests (`tests/`, 18 tests, `pytest -q`): `test_data.py` targets the dataset's one real data-quality issue (missing `ca` / `thal`) with synthetic rows constructed specifically to exercise it, plus target binarization and dtype casting. `test_features.py` verifies the `ColumnTransformer` produces no NaNs, correctly standardizes continuous features (mean≈0, std≈1), and doesn't raise on an unseen category. `test_model.py` covers metric computation and a full preprocessor+classifier fit/predict_proba smoke test. `test_api.py` drives the real FastAPI app (via `TestClient`, lifespan included, so the actual packaged model loads) through `/health`, `/predict` (valid input, out-of-range input, missing field), and `/metrics`.

Manual API testing beyond `pytest : curl` (used throughout every verification in this report), and FastAPI's auto-generated Swagger UI at `/docs`, confirmed live (not just assumed from the framework default) — `/docs` returns 200 and `/openapi.json` correctly enumerates all three endpoints and their `PatientFeatures` / `PredictionResponse` schemas:

Heart Disease Risk Classifier API 1.0.0 OAS 3.1
/openapi.json
Predicts heart disease presence/absence from patient health data.

default

- GET** /health Health
- GET** /metrics Metrics
- POST** /predict Predict

Schemas

- HTTPValidationError** > Expand all object
- PatientFeatures** > Expand all object
- PredictionResponse** > Expand all object
- ValidationError** > Expand all object

`.github/workflows/ci.yml` — three jobs, each failing loudly on any error (no `--no-verify` -style bypasses, `curl -f` rather than bare `curl` so a broken health check actually fails the job):

1. **lint** — `ruff check .` + `black --check .`.
2. **test** (needs lint) — downloads the dataset, preprocesses, runs EDA, trains with `--fast`, then `pytest` with coverage; uploads test results, EDA figures, and the fast-trained model as workflow artifacts.
3. **docker** (needs test) — downloads the model artifact from the test job, `docker build` s the image, runs it, polls `/health` until it passes (or fails the job after 30s), then POSTs `sample_input.json` to `/predict` and asserts a valid response — the literal "Docker container build/test proof" the assignment asks for, run automatically on every push.

Every command each job runs was validated locally end-to-end first (lint, `pytest --cov`, `train.py --fast`, and the exact `docker build` / `docker run` / `health-poll-loop` / `/predict` smoke test used in the `docker` job), then confirmed on GitHub's own runners after pushing: two pushes, two green runs, all three jobs passing both times — [run #1](#) and [run #2](#), queried directly via the GitHub API (`lint` : success, `test` : success, `docker` : success, on both). Nothing environment-specific broke between local validation and the hosted runner.

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Workflow Event Status Branch Actor

Fill in repository link now that the repo is pushed to GitHub

CI #2: Commit cd7085d pushed by muthukrishnanram

main

7 minutes ago

5m 22s

...

Verify Swagger UI live and fix a duplicated report section

CI #1: Commit 9b78841 pushed by muthukrishnanram

main

13 minutes ago

5m 41s

...

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Fill in repository link now that the repo is pushed to GitHub #2

Sign in to view logs

Summary

All jobs

lint test docker

Run details

Usage

Workflow file

Triggered via push 7 minutes ago

muthukrishnanram pushed cd7085d main

Status

Success

Total duration

5m 22s

Artifacts

3

ci.yml

on: push

lint 1m 11s test 2m 44s docker 1m 19s

Annotations

3 warnings

lint

Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, ...

Show more

test

Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, ...

Show more

docker

Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, ...

Show more

9. Model Containerization

`Dockerfile` — multi-stage build: a `builder` stage installs `requirements.txt` into `/root/.local`, and the final stage (also `python:3.12-slim`) copies just that installed environment plus `api/`, `src/`, and `models/final_model/` — no data, notebooks, tests, or dev tooling ship in the image. Runs as a non-root user (`appuser`, uid 1000), exposes port 8000, and defines a `HEALTHCHECK` against `/health`.

```
docker build -t heart-disease-api .
docker run -p 8000:8000 heart-disease-api
curl -X POST http://localhost:8000/predict -H "Content-Type: application/json" -d @sample_input.json
```

Verified locally (full transcript in `screenshots/docker_build_run_log.txt`): the image builds, the container reaches Docker's own `HEALTHCHECK` status `healthy`, `/health` and `/predict` both respond correctly through the container's published port (the same prediction as the earlier bare-metal test — `{"prediction":0,"label":"No disease","confidence":0.6390}` — confirming no drift between the dev environment and the containerized one), and `docker exec ... id` confirms the process runs as `appuser` (uid 1000), not root. Final image size: 623 MB content (up from 219 MB before adding XGBoost — its native shared library is the single largest dependency in the image; still an acceptable tradeoff for a third model that meaningfully adds to the comparison in Section 5).

10. Production Deployment (Kubernetes / Minikube)

`k8s/deployment.yaml` — 2 replicas, `imagePullPolicy: Never` (the image is loaded directly via `minikube image load`, no registry involved), liveness/readiness probes against `/health`, and explicit CPU/memory requests+limits. `k8s/service.yaml` exposes it as a `LoadBalancer`; `k8s/ingress.yaml` is documented as a `sudo` -tunnel-free alternative. Full commands in `k8s/README.md`:

```
minikube start --driver=docker --cpus=4 --memory=8192
docker build -t heart-disease-api:latest .
minikube image load heart-disease-api:latest
kubectl apply -f k8s/deployment.yaml -f k8s/service.yaml
minikube tunnel # separate terminal - populates the LoadBalancer's EXTERNAL-IP
kubectl get svc heart-disease-api
curl http://<EXTERNAL-IP>/health
```

Verified locally (full transcript in `screenshots/minikube_deployment_log.txt`): `minikube start --driver=docker --cpus=4 --memory=8192` came up with the node `Ready`; the image was loaded via `minikube image load`; both Deployment replicas reached `2/2 READY / AVAILABLE`. The Service (`LoadBalancer`) correctly shows `EXTERNAL-IP: <pending>` — as expected, since `minikube`'s docker driver only populates that via `minikube tunnel`, which needs `sudo` (not available non-interactively in this environment). Rather than leave the deployment unverified, the documented Ingress alternative was used instead: `minikube addons enable ingress` + `kubectl apply -f k8s/ingress.yaml` brought up `ingress-nginx`, and `curl --resolve heart-api.local:80:$(minikube ip) http://heart-api.local/...` against both `/health` and `/predict` returned correct responses — the same prediction as the bare-metal and Docker checks,

now served from inside a 2-replica Kubernetes deployment with working liveness/readiness probes. `minikube tunnel` remains documented in `k8s/README.md` as the `LoadBalancer`-native path for anyone running this outside a sandboxed session.

Re-verified after adding XGBoost and rebuilding the image: `minikube image load` + `kubectl rollout restart deployment/heart-disease-api` rolled both replicas over to the new image with zero manual pod deletion, and the same `curl` checks against `/health` and `/predict` returned the identical prediction as before — expected, since Random Forest (the deployed model) didn't change; only the image's dependency set did.

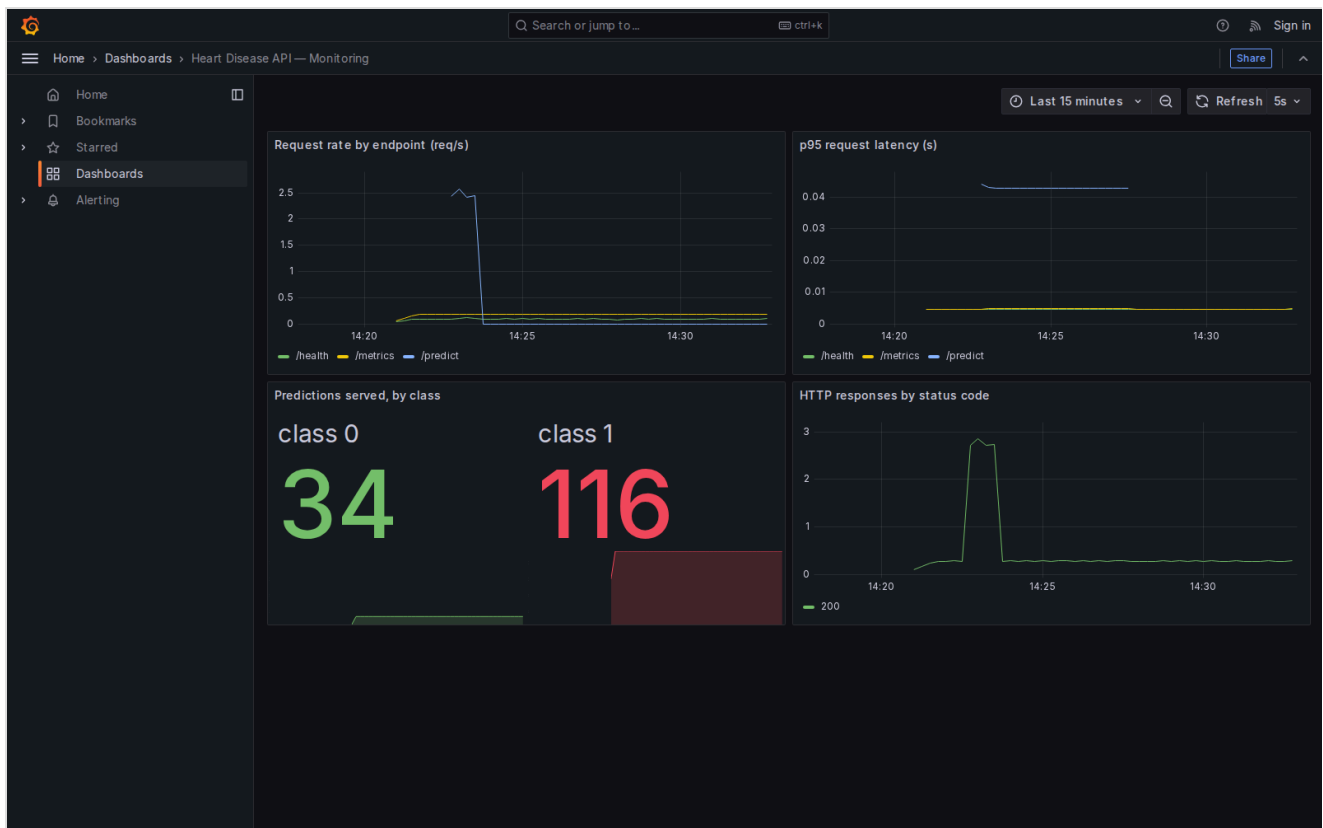
11. Monitoring & Logging

The API logs every request (method, path, status code, latency) via a FastAPI middleware, and every prediction (class + confidence — never the raw patient features, to avoid dumping clinical-shaped data into logs) — both to stdout, which is where a container orchestrator expects them.

`/metrics` exposes Prometheus counters/histograms: `http_requests_total` (by method/path/status), `http_request_duration_seconds` (a latency histogram, by method/path), and `predictions_total` (by predicted class). `docker-compose.yml` stands up the API alongside Prometheus (scraping `/metrics` every 5s) and Grafana (provisioned with a datasource and a 4-panel dashboard: request rate, p95 latency, predictions by class, status codes) with zero manual clicking required:

```
docker compose up
python scripts/generate_load.py --n 200 # populate the dashboard with real traffic
```

Verified locally (full transcript in `screenshots/monitoring_stack_log.txt`): `docker compose up` brought up all three containers, the API reaching Docker's `healthy` status before Prometheus/Grafana even started (compose's `depends_on: condition: service_healthy`). Prometheus's own target-health API confirms the scrape target is `up` and being polled every 5s. `scripts/generate_load.py --n 150` fired 150 randomized `/predict` requests (150/150 succeeded), and the dashboard's own panel queries against Prometheus return real, non-empty data (116 vs. 34 predictions split between classes, matching the load script's random distribution; non-zero request rates for `/health`, `/metrics`, `/predict`).



The screenshot above is a real capture of the live dashboard (via a headless-Chromium container, since installing a browser directly in this sandboxed environment would have needed `sudo`) — note the request-rate spike from the load-generation burst, the 34/116 prediction split, and the HTTP 200 status-code timeseries, all sourced from the same `/metrics` counters `api/main.py` increments per request.

12. Conclusion

The pipeline covers the full assignment scope end-to-end, and every stage was actually run and verified rather than assumed — not just files that look right: EDA and model training execute from a clean venv; three cross-validated, MLflow-tracked models (registered in the Model Registry) produced a packaged artifact that was proven portable by loading it in a throwaway serving-only venv (which caught and fixed two real bugs — a missing dependency and an untrusted-type serialization error); 18 unit/API tests pass; the Docker image builds, runs, and passes its own `HEALTHCHECK` as a non-root user; the same image deploys cleanly to a 2-replica Minikube deployment reachable via Ingress; and the Prometheus/Grafana stack shows a live dashboard populated by real generated traffic.

The repository is public at <https://github.com/muthukrishnanram/mlops-assignment1>. `ci.yml` has run twice on GitHub's own runners so far, both green across all three jobs — the last outstanding "should work" in this report is now a confirmed "did work," closing the loop between local validation and the actual hosted pipeline.

Appendix: Verification Artifacts

Every claim above that isn't visible in the report's own figures is backed by a transcript in `screenshots/` :

- `docker_build_run_log.txt` — Docker build, run, health-check, non-root check.
- `minikube_deployment_log.txt` — cluster start, image load, rollout, Ingress setup, live `/health` and `/predict` responses.
- `monitoring_stack_log.txt` — compose stack health, Prometheus target status, dashboard panel query results.
- `grafana_dashboard.png` — live screenshot of the populated dashboard.

Appendix: Repository Layout

See `README.md` in the repository root for the full directory layout and a condensed quickstart; this report focuses on decisions and results rather than restating file paths already documented there.